

ESP Production Drop

Field Manual — Setup and Troubleshooting Reference

Use case definition. An electric submersible pump lifts well fluids through a downhole motor, protector, pump stages, tubing, and surface flowline. Abnormal behavior includes falling BOPD/BLPD, unstable current, pressure changes, gas interference, or a mismatch between well tests and instrumentation. Causes include inflow decline, gas entry, frequency changes, surface restrictions, power interruption, and faulty transmitters. Troubleshooting protects the motor, cable, pump stages, and well integrity.

Manual setup and use. Use the well operating envelope, ESP vendor curves, approved restart logic, and OEM/site alarm or trip limits. The values are training/reference values.

Safety note. Keep people clear of wellhead and electrical hazards; do not change frequency, open wellhead equipment, or manipulate valves outside the approved well procedure.

Quick-use workflow

Use this sequence for every event: establish a safe operating state; capture the initial numerical readings; compare each value with the stated reference; apply the trigger-based action; correct the identified component or process path; restore operation gradually; and confirm the expected numerical result. If an OEM/site alarm or trip limit differs from a training value, the OEM/site limit governs.

Scenario 1: Sudden production decline

Field Condition

Oil 620 BOPD from 910; total liquid 1,420 BLPD from 1,880; tubing 74 barg; casing 18 barg; 52 Hz

Problem

Production dropped 32 % with stable frequency, suggesting gas interference, inflow change, or a surface restriction.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses oil below 700 BOPD.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when stabilize 850–950 BOPD and 1,750–1,950 BLPD.

Expected Result

stabilize 850–950 BOPD and 1,750–1,950 BLPD

Scenario 2: Gradual production decline

Field Condition

Oil 540 BOPD from 720 over 14 days; water cut 68 % from 55 %; tubing 82 barg; 50 Hz

Problem

A 25 % decline with rising water cut indicates changing inflow or pump wear.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses oil below 600 BOPD.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.

4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when trend returns above 650 BOPD or approved decline diagnosis is confirmed.

Expected Result

trend returns above 650 BOPD or approved decline diagnosis is confirmed

Scenario 3: ESP underload

Field Condition

motor current 24 A (normal 38–45); 48 Hz; tubing 61 barg; oil 410 BOPD

Problem

Low current and low liquid rate indicate reduced pump loading or gas interference.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses current below 30 A.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when current 38–45 A with stable production.

Expected Result

current 38–45 A with stable production

Scenario 4: ESP overload

Field Condition

current 59 A (normal 38–45); voltage 2,380 V; 52 Hz; tubing 96 barg; oil 780 BOPD

Problem

High current suggests excessive head, solids, or pump/mechanical loading.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses current above 52 A.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when current 38–45 A and tubing near 75–90 barg.

Expected Result

current 38–45 A and tubing near 75–90 barg

Scenario 5: Motor current instability

Field Condition

current oscillates 31–49 A every 18 s; voltage 2,360–2,420 V; 50 Hz

Problem

Electrical supply instability or intermittent gas loading is stressing the ESP.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses swing exceeds 10 A.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when current variation below 5 A.

Expected Result

current variation below 5 A

Scenario 6: Frequency-related performance problem

Field Condition

frequency 42 Hz versus 52 Hz reference; oil 430 BOPD versus 860; tubing 58 barg

Problem

The commanded frequency is limiting pump speed and production.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses frequency below approved target by >3 Hz.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when frequency restored to approved 50–52 Hz and oil near 800–900 BOPD.

Expected Result

frequency restored to approved 50–52 Hz and oil near 800–900 BOPD

Scenario 7: Gas interference

Field Condition

gas rate 1.8 MMscf/d from 0.9; current 27–46 A; oil 390 BOPD; casing 24 barg

Problem

High free gas is causing unstable pump intake performance.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses current swing above 8 A.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when stable current 38–45 A and oil above 650 BOPD.

Expected Result

stable current 38–45 A and oil above 650 BOPD

Scenario 8: Reduced well inflow

Field Condition

tubing 69 barg; pump discharge 98 barg; oil 280 BOPD; water cut 72 %

Problem

High pump discharge with low oil indicates inflow limitation rather than a surface flowline issue.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses oil below 350 BOPD.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when approved drawdown/inflow test completed with stable intake conditions.

Expected Result

approved drawdown/inflow test completed with stable intake conditions

Scenario 9: High casing pressure

Field Condition

casing 38 barg (normal 15–22); tubing 88 barg; oil 510 BOPD

Problem

Elevated casing pressure reduces drawdown and may indicate annular restriction or gas handling issue.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses casing above 30 barg.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when casing 15–22 barg and production stable.

Expected Result

casing 15–22 barg and production stable

Scenario 10: Low discharge pressure

Field Condition

pump discharge 54 barg (normal 85–105); current 29 A; oil 360 BOPD

Problem

Low discharge pressure and underload indicate loss of pump head, gas lock, or sensor error.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses pressure below 70 barg.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.

3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when discharge 85–105 barg and current 38–45 A.

Expected Result

discharge 85–105 barg and current 38–45 A

Scenario 11: Flowline restriction

Field Condition

tubing 112 barg; downstream 68 barg; oil 460 BOPD from 820; choke 72 %

Problem

High upstream/downstream differential indicates a restricted flowline or surface choke path.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses differential above 35 barg.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when differential below 20 barg and oil above 700 BOPD.

Expected Result

differential below 20 barg and oil above 700 BOPD

Scenario 12: Surface valve restriction

Field Condition

wellhead choke 18 % versus 55 %; tubing 106 barg; oil 390 BOPD

Problem

A restricted choke setting is limiting surface throughput.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses choke below 25 % without approved reason.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when choke 45–60 % with tubing 75–90 barg.

Expected Result

choke 45–60 % with tubing 75–90 barg

Scenario 13: Power interruption

Field Condition

ESP stopped at 03:14; voltage fell to 0 V for 4 s; restart oil 120 BOPD; current 19 A

Problem

A power event followed by low-load restart has not restored pump performance.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses voltage interruption >1 s.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when approved restart completed with current 38–45 A.

Expected Result

approved restart completed with current 38–45 A

Scenario 14: Restart with low production

Field Condition

restart at 50 Hz; oil 210 BOPD; tubing 62 barg; current 25 A; casing 26 barg

Problem

The pump restarted but remains underloaded, consistent with gas lock or insufficient inflow.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses oil below 300 BOPD after 30 min.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when oil above 600 BOPD or approved shutdown/restart decision.

Expected Result

oil above 600 BOPD or approved shutdown/restart decision

Scenario 15: Instrumentation error

Field Condition

SCADA oil 390 BOPD; test separator 760 BOPD; tubing transmitters differ by 9 barg

Problem

A large measurement disagreement may be falsely indicating production loss.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses flow disagreement above 10 %.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when indications within 5 % after calibration.

Expected Result

indications within 5 % after calibration